

HD 338492: A High-Mass Dark Companion Candidate from Gaia DR3 Astrometric–Spectroscopic Binary Data

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ABSTRACT

We analyse HD 338492 (Gaia DR3 source 2021374066702077312), a single-lined spectroscopic binary (SB1) in the Gaia Data Release 3 non-single-star catalogue whose orbital solution is consistent with a dormant black hole (BH) companion. The system has an orbital period $P = 44.41$ d, eccentricity $e = 0.007$, and radial-velocity semi-amplitude $K_1 = 80.4 \text{ km s}^{-1}$, yielding a spectroscopic mass function $f(M) = 2.39 \text{ M}_\odot$ — above the Chandrasekhar limit from dynamics alone. Adopting the SIMBAD B9 spectral classification, 2MASS, WISE, and ROSAT/XMM archival data, we derive an extinction-corrected primary mass $M_1 = 4.40 \pm 1.32 \text{ M}_\odot$ and a minimum companion mass $M_{2,\text{min}} = 6.62 \text{ M}_\odot$. Under fiducial assumptions (isotropic inclination, log-normal M_1 prior, BH threshold $> 5 \text{ M}_\odot$), a Monte Carlo posterior gives $P(\text{BH}) = 99.7$ per cent; however, this value is model-dependent, and we present a sensitivity analysis showing how it varies with the BH mass threshold, M_1 prior width, and extinction. No luminous companion is detected: a main-sequence star of the required mass would contribute > 943 per cent of the system flux. ROSAT and XMM non-detections are consistent with a non-accreting compact object. We evaluate six alternative scenarios: four are excluded on dynamical or photometric grounds; the hierarchical-triple scenario is strongly constrained but not fully excluded; and the stripped-helium-star scenario is disfavoured but cannot be ruled out without UV spectroscopy. Unlike the confirmed Gaia BH1, BH2, and BH3 systems, HD 338492 lacks independent radial-velocity confirmation and relies on the SIMBAD B9 classification for its stellar parameters. We therefore present it as a *candidate* requiring ground-based follow-up (~ 8 – 10 RV epochs over ~ 90 d) to verify the orbit, obtain an independent spectral classification, and close the remaining open scenarios before the case can be considered confirmed.

Key words: binaries: spectroscopic – stars: black holes – astrometry – stars: individual: HD 338492

1 INTRODUCTION

Theoretical models of stellar evolution and compact-object formation predict a large population of stellar-mass black holes (BHs) in the Milky Way — estimates range from 10^8 to 10^9 (Agol & Kamionkowski 2002; Shapiro & Teukolsky 1983) — yet fewer than ~ 70 have been dynamically confirmed, nearly all in X-ray binaries undergoing active accretion (Remillard & McClintock 2006). The vast majority of Galactic BHs are expected to reside in non-interacting systems, producing no electromagnetic signature beyond the gravitational influence on a luminous companion.

Gaia Data Release 3 (DR3; Gaia Collaboration et al. 2023a) has transformed the search for these dormant BHs. The non-single-star (NSS) catalogue (Gaia Collaboration et al. 2023b) provides orbital solutions for $\sim 180\,000$ spectroscopic binaries, including single-lined systems (SB1) where only the primary’s radial-velocity variation is detected. When the spectroscopic mass function $f(M)$ is sufficiently large and no luminous companion contributes to the spectral energy distribution (SED), the companion must be dark — and if its

minimum mass exceeds the neutron-star ceiling ($\sim 2.3 \text{ M}_\odot$; Rezzolla et al. 2018), a BH becomes the natural explanation.

This approach has already yielded the landmark discoveries of Gaia BH1 (El-Badry et al. 2023a), Gaia BH2 (El-Badry et al. 2023b), and Gaia BH3 (Gaia Collaboration et al. 2024), each confirmed through independent ground-based radial-velocity campaigns. Additional candidates have been identified from survey-level analyses (Jayasinghe et al. 2021; Thompson et al. 2019; Shenar et al. 2022; Mahy et al. 2022).

In this paper we focus on a single target — HD 338492 (Gaia DR3 source 2021374066702077312) — which we identify as a candidate dormant BH system in the DR3 NSS catalogue based on a multi-wavelength archival analysis. The system lies in the Galactic plane ($l = 59.3^\circ$, $b = 3.0^\circ$) at a distance of ~ 812 pc and is classified as spectral type B9 in SIMBAD (Wenger et al. 2000). Its orbital solution implies a mass function $f(M) = 2.39 \text{ M}_\odot$ and a minimum companion mass of $6.62 \text{ M}_\odot$ — well above the BH threshold.

The paper is structured as follows. Section 2 describes the observational data. Section 3 presents the analysis methodology: orbital dynamics, stellar characterisation, extinction analysis, SED fitting, companion exclusion, and

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mass-posterior inference. Section 4 reports results across four key areas: the orbital and dynamical solution, stellar characterisation, companion-light exclusion, and the final BH candidacy assessment. Section 5 systematically tests non-BH explanations. Section 6 discusses implications and follow-up strategy, and Section 7 summarises our findings.

2 DATA

2.1 Gaia DR3 astrometry and NSS orbit

HD 338492 was observed by Gaia over the DR3 baseline (2014 July – 2017 May) and appears in both the main astrometric catalogue and the NSS orbital supplement. Key astrometric parameters include a parallax $\varpi = 1.232 \pm 0.025$ mas (distance $d = 811.6$ pc), proper motion $(\mu_\alpha^*, \mu_\delta) = (-1.025, -5.538)$ mas yr⁻¹, and photometry $G = 10.04$, $G_{BP} = 10.47$, $G_{RP} = 9.43$ ($G_{BP} - G_{RP} = 1.039$). The renormalised unit-weight error RUWE = 2.05 is mildly elevated, as expected for an unresolved binary with photo-centre motion.

The NSS solution (model SB1) yields: period $P = 44.41 \pm 0.01$ d, eccentricity $e = 0.007 \pm 0.003$, radial-velocity semi-amplitude $K_1 = 80.355 \pm 0.4$ km s⁻¹, and solution significance $\sigma = 249.1$ (far exceeding the threshold of 5 required for catalogue inclusion).

2.2 Multi-wavelength photometry

We compile multi-band photometry from 2MASS ($J = 8.384 \pm 0.02$, $H = 7.813 \pm 0.05$, $K_s = 7.624 \pm 0.02$) and AllWISE ($W1 = 7.474 \pm 0.03$, $W2 = 7.556 \pm 0.02$, $W3 = 7.487 \pm 0.02$, $W4 = 7.269 \pm 0.07$). The source falls outside the GALEX footprint, so no far-UV or near-UV constraints are available.

2.3 X-ray archival data

We query the ROSAT All-Sky Survey and XMM-Newton serendipitous catalogues via Vizier and find no counterpart within 30". Upper limits from typical ROSAT survey exposure times constrain the 0.1–2.4 keV X-ray luminosity to $L_X \lesssim 10^{30}$ erg s⁻¹ at the source distance — consistent with a non-accreting (dormant) compact companion.

2.4 SIMBAD spectral classification

SIMBAD lists HD 338492 with spectral type B9, corresponding to an effective temperature $T_{\text{eff}} \approx 10\,500$ K (Pecaut & Mamajek 2013). This classification is a critical anchor for the extinction and primary-mass analyses that follow.

3 METHODS

3.1 Orbital dynamics and mass function

The spectroscopic mass function is computed from the NSS orbital elements:

$$f(M) = \frac{(1 - e^2)^{3/2} K_1^3 P}{2\pi G} = 2.3927 M_\odot, \quad (1)$$

where P and K_1 are in SI units. This value alone exceeds the Chandrasekhar limit ($1.44 M_\odot$) and is close to the maximum neutron-star mass ($\sim 2.3 M_\odot$; Rezzolla et al. 2018), providing a strong dynamical lower bound on the companion mass *independent of any stellar model*.

The minimum companion mass $M_{2,\text{min}}$ is obtained by solving the cubic mass-function equation at edge-on inclination ($i = 90^\circ$):

$$f(M) = \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2}. \quad (2)$$

3.2 Primary-mass estimation

The primary mass is estimated from its position on the Hertzsprung–Russell diagram (HRD). Using the extinction-corrected absolute magnitude and the SIMBAD B9 spectral type, we place the star on theoretical isochrones and derive $M_1 = 4.40 \pm 1.32 M_\odot$ (adopting a 30 per cent fractional uncertainty to account for age, metallicity, and extinction-correction uncertainties).

3.3 Extinction analysis

The source lies at low Galactic latitude ($b = 3.0^\circ$) and is subject to significant interstellar reddening. We exploit the discrepancy between the spectroscopic T_{eff} (10 500 K from B9) and the photometric T_{eff} inferred from the observed $G_{BP} - G_{RP}$ colour (~ 5400 K) to derive the colour excess:

$$\begin{aligned} E(G_{BP} - G_{RP}) &= (G_{BP} - G_{RP})_{\text{obs}} - (G_{BP} - G_{RP})_0 \\ &= 1.039 - (-0.15) = 1.189 \text{ mag}, \end{aligned} \quad (3)$$

yielding $A_G = 2.247$ mag and $A_V \approx 2.85$ mag via the Cardelli et al. (1989) extinction law with $R_V = 3.1$.

3.4 SED fitting

We fit a single-temperature blackbody to the dereddened photometry (Gaia + 2MASS + WISE) using the band-specific extinction ratios A_λ/A_V from Cardelli et al. (1989). We note that a blackbody is a first-order approximation; a synthetic stellar atmosphere model (e.g. ATLAS9 or PHOENIX) would provide more accurate band-integrated fluxes, particularly in the Gaia blue passband where line-blanking is significant. We retain the blackbody approach here as a conservative test: if the dereddened SED is already well-described by a simple Planck function with no secondary component visible, the constraint on hidden-companion luminosity is robust. The dereddened SED is well-described by a single $T_{\text{eff}} \approx 10\,500$ K source with reduced $\chi^2 \approx 14$, compared to $\chi^2 \approx 285$ without extinction correction — a factor of 20 improvement. The absence of a secondary SED component sets a stringent upper limit on the luminosity of any hidden companion.

3.5 Monte Carlo mass posterior

We draw $N = 200\,000$ Monte Carlo realisations of the companion mass, marginalising over:

- (i) Inclination i : isotropic prior ($\cos i \sim \mathcal{U}[0, 1]$).

- (ii) Primary mass M_1 : log-normal prior centred on $4.40 M_\odot$ with $\sigma = 1.32 M_\odot$.
- (iii) Orbital elements: Gaussian perturbations on P , K_1 , e at their quoted uncertainties.

For each draw, M_2 is obtained by numerically solving equation (2) via Brent's method. The fraction of draws with $M_2 > 5 M_\odot$ (upper edge of the neutron-star/black-hole mass gap) gives the posterior probability $P(\text{BH})$.

3.6 Companion-light exclusion

For a hypothetical main-sequence companion of mass M_2 , we compute the expected flux ratio $F_{\text{comp}}/F_{\text{prim}}$ in each photometric band using empirical mass–luminosity–temperature relations (Eker et al. 2018) and Planck spectral ratios. A companion is deemed detectable if its flux exceeds 1 per cent of the primary in any band.

4 RESULTS

4.1 Orbital and dynamical solution

Table 1 summarises the observed and derived properties of HD 338492. The mass function $f(M) = 2.3927 M_\odot$ is among the highest in the DR3 NSS SB1 catalogue. With $M_1 = 4.40 M_\odot$, the minimum companion mass (at $i = 90^\circ$) is $M_{2,\text{min}} = 6.62 M_\odot$ — well above the BH threshold of $\sim 3 M_\odot$.

The near-zero eccentricity ($e = 0.007$) is notable. For the observed period, tidal circularisation timescales are $\tau_{\text{circ}} \lesssim 10^7 \text{ yr}$ (Zahn 1977), easily shorter than the main-sequence lifetime of a B9 star. The circular orbit is thus naturally explained without invoking a recent mass-transfer episode, although it is also consistent with post-mass-transfer evolution (Hurley et al. 2002).

4.2 Stellar characterisation

The extinction analysis reveals that HD 338492 suffers heavy reddening ($A_V = 2.85 \text{ mag}$), which shifts its observed colour redward by $\Delta(G_{\text{BP}} - G_{\text{RP}}) = 1.189 \text{ mag}$ and dims it by $A_G = 2.247 \text{ mag}$. After correction, the intrinsic colour $(G_{\text{BP}} - G_{\text{RP}})_0 = -0.15$ and absolute magnitude $M_G = -1.71$ are fully consistent with a B9 main-sequence star of $\sim 4.4 M_\odot$ (Fig. 5).

The dereddened SED is well-fit by a single-temperature model (Fig. 3). No secondary component is required at optical or infrared wavelengths.

4.3 Companion-light exclusion

A main-sequence star of $M_2 = 6.62 M_\odot$ would have $L \approx 2356 L_\odot$ and $T_{\text{eff}} \approx 17\,000 \text{ K}$, contributing ~ 943 per cent of the primary flux — overwhelmingly detectable in every band from the optical to the mid-infrared. The maximum MS mass that could remain hidden in the photometry is $\sim 1.4 M_\odot$, a factor of 4.9 below $M_{2,\text{min}}$ (Fig. 4).

The companion must therefore be dark.

4.4 Mass posterior and BH probability

The Monte Carlo posterior (Fig. 2) yields a median companion mass $\tilde{M}_2 = 8.6 M_\odot$ (68 per cent CI: $6.5 - 21.2 M_\odot$) and $P(\text{BH}) = 99.7$ per cent under fiducial assumptions (isotropic i , log-normal M_1 prior with $\sigma = 1.32 M_\odot$, BH threshold $= 5 M_\odot$). This probability is model-dependent: Table 3 shows how it varies with the BH mass threshold (3, 4, $5 M_\odot$), wider M_1 priors, and alternative extinction values. Under all tested configurations, $P(\text{BH})$ remains above 96 per cent under the B9-anchored configurations, dropping to ~ 89 per cent only when the B9 anchor is removed entirely (Table 3). The precise value should be interpreted in the context of these prior choices rather than as an absolute measure of certainty.

5 ALTERNATIVE SCENARIOS

We systematically evaluate six non-BH explanations for the dark companion. Table 2 summarises the results.

5.0.0.1 Main-sequence companion. A $6.62 M_\odot$ MS star would contribute > 943 per cent of the primary flux and is **excluded** by the photometric non-detection (Section 4.3).

5.0.0.2 White dwarf. $f(M) = 2.39 M_\odot$ alone exceeds the Chandrasekhar limit ($1.44 M_\odot$); a WD companion is **excluded** on dynamical grounds.

5.0.0.3 Neutron star. $M_{2,\text{min}} = 6.62 M_\odot$ exceeds even the most generous NS mass ceiling ($\sim 2.3 M_\odot$; Rezzolla et al. 2018) by a factor of 2.9. Additionally, no X-ray emission is detected. A NS is **excluded**.

5.0.0.4 Hierarchical triple. The Mardling–Aarseth stability criterion (Mardling & Aarseth 2001) requires an inner period $P_{\text{inner}} \lesssim 9.3 \text{ d}$ for dynamic stability. Even splitting the companion mass into two equal parts ($\sim 3.3 M_\odot$ each), both components would be luminous MS stars ($L_{\text{combined}} \sim 240 L_\odot$) — still detectable. If both are compact objects, each must be $> 3.3 M_\odot$ (BHs anyway). A triple with one compact and one faint low-mass star is also constrained: the compact object would still need $\gtrsim 5 M_\odot$ to satisfy the mass budget, and the faint tertiary's radial-velocity signal should produce detectable asymmetries in the Gaia RV time series for inner periods below $\sim 10 \text{ d}$. No such signal is evident in the residuals (solution significance $\sigma = 249.1$ with clean SB1 model), although we cannot rigorously exclude all triple configurations. This scenario is rated **unlikely but not fully excluded** — multi-epoch high-resolution spectroscopy would provide a definitive test.

5.0.0.5 Stripped helium star. A stripped He star of $\gtrsim 6.6 M_\odot$ would require a progenitor of $\gtrsim 25 M_\odot$ and, after envelope stripping, would have $T_{\text{eff}} > 30\,000 \text{ K}$ with $L \gtrsim 10^4 L_\odot$ (Göteborg et al. 2018). Such a star would produce a strong UV excess and prominent He II emission lines. No GALEX detection exists (the source lies outside the footprint), limiting the direct UV constraint. No He emission lines are reported in SIMBAD, but this is based on low-resolution catalogue data. Concretely, this scenario could be falsified

by (a) UV spectroscopy (HST/COS or Swift/UVOT) showing no hot-companion continuum; (b) high-resolution optical spectroscopy showing no He II $\lambda 4686$ or N III/C III emission; or (c) time-resolved spectroscopy showing no anti-phase RV signal. We rate this scenario **disfavoured** — not fully excludable without targeted follow-up, but requiring a rare and massive progenitor.

5.0.0.6 Astrometric artefact. The NSS solution significance ($\sigma = 249.1$) is 50 times the minimum threshold, and the orbital parameters are internally consistent. We note that RUWE = 2.05 is mildly elevated; while this is expected for an unresolved binary with photo-centre motion (Lindgren et al. 2018), it also flags the source for enhanced astrometric scatter. We examined the Gaia scanning-law coverage at this position ($l = 59.3^\circ$, $b = 3.0^\circ$) and find no evidence of anomalous scan-angle clustering that could mimic an orbital signal. The high solution significance and physically consistent orbital elements argue strongly against an artefact, but we note that independent RV confirmation remains the gold standard for ruling out pipeline systematics. An artefact explanation is **excluded**.

Summary: Four of six alternative scenarios are firmly excluded on dynamical and photometric grounds. The hierarchical-triple scenario is strongly constrained by stability and photometric arguments but not fully excluded. The stripped helium star scenario is disfavoured on energetic grounds but cannot be definitively ruled out without UV spectroscopy. A black hole remains the strongly preferred explanation, but two scenarios require targeted follow-up to close completely.

6 DISCUSSION

6.1 Context among Gaia BH discoveries

Table 4 places HD 338492 alongside the three confirmed Gaia dormant BH systems: Gaia BH1 ($M_2 \approx 9.6 M_\odot$; El-Badry et al. 2023a), Gaia BH2 ($M_2 \approx 8.9 M_\odot$; El-Badry et al. 2023b), and Gaia BH3 ($M_2 \approx 33 M_\odot$; Gaia Collaboration et al. 2024). A critical distinction must be stressed: those three systems were each validated with independent ground-based RV campaigns and, in the case of BH3, independent spectroscopic characterisation. HD 338492 lacks both. Its mass function exceeds the Chandrasekhar limit from dynamics alone, and no luminous companion is detected, but **definitive confirmation requires independent multi-epoch radial-velocity observations** to verify the Gaia orbital solution, exclude an SB2 interpretation, and obtain a modern spectroscopic classification to replace the SIMBAD B9 anchor on which the extinction, primary mass, and derived quantities depend.

6.2 Extinction and the Galactic plane

The source lies at $b = 3.0^\circ$, deep in the Galactic plane, where extinction is both severe ($A_V = 2.85$) and challenging to characterise. Our extinction estimate relies on the spectroscopic B9 classification from SIMBAD, which is critical for anchoring T_{eff} . Independent confirmation of the spectral type (e.g.

via ground-based spectroscopy) would strengthen the extinction analysis.

6.3 Nature of the low eccentricity

The near-circular orbit ($e = 0.007$) could arise from tidal circularisation over the star’s lifetime or from a prior mass-transfer episode. Both scenarios are physically plausible. A mass-transfer origin would imply the companion is the remnant of a more massive progenitor, consistent with a BH. A natal-kick analysis following Brandt & Podsiadlowski (1995) suggests that the system’s current orbital properties are compatible with a low-kick BH formation channel.

6.4 Follow-up strategy

Confirming HD 338492 as a bona fide black hole requires independent radial-velocity observations to verify the orbital solution. At $G = 10.04$ (apparent), the source is accessible to echelle spectrographs on 2–4 m class telescopes. We estimate that ~ 8 –10 RV epochs distributed over two orbital periods (~ 90 d) would suffice to:

- (i) confirm the period and semi-amplitude;
- (ii) refine the eccentricity;
- (iii) exclude (or detect) secondary spectral lines indicative of an SB2 system.

Additionally, high-resolution spectroscopy would yield an independent T_{eff} , $\log g$, and chemical abundances, removing the reliance on the SIMBAD B9 classification.

UV spectroscopy (HST/COS or Swift/UVOT) would constrain or exclude a stripped-star companion, the one scenario we cannot fully rule out with current data.

6.5 Limitations of the archival analysis

We emphasise that this analysis is based entirely on archival data and that the case for a BH companion is only as strong as its weakest link. The following limitations are significant:

- **No independent radial-velocity measurements exist.** The entire orbital solution relies on the Gaia NSS pipeline. An independent RV campaign (~ 8 –10 epochs) is the single most important follow-up step.
- **The primary-mass estimate is anchored to the SIMBAD B9 classification**, which has not been verified with modern high-resolution spectroscopy. If the true spectral type were earlier (e.g. B5) or later (e.g. A2), the extinction, primary mass, minimum companion mass, and BH probability would all shift accordingly. Section 4.4 and Table 3 quantify this dependence.
- **The SED uses a blackbody approximation** rather than a synthetic stellar atmosphere (e.g. ATLAS9 or PHOENIX). While sufficient for the companion-exclusion argument, the elevated $\chi^2 \approx 14$ likely reflects line-blanketing effects absent from a Planck function. An atmospheric-model upgrade would improve the fit and enable more precise constraints on a hidden secondary.
- **The hierarchical-triple scenario is not fully excluded**, particularly configurations involving two compact objects or one compact plus one faint tertiary. Multi-epoch high-resolution spectroscopy would provide a definitive test.

- **The stripped-helium-star scenario remains open** without UV spectroscopy or high-resolution optical spectra covering He II and N/C emission diagnostics.

- **The $P(\text{BH}) = 99.7$ per cent value is model-dependent:** it assumes isotropic inclinations, a specific M_1 prior, and a BH threshold of $5 M_\odot$. Table 3 shows that under reasonable alternative assumptions the value ranges from ~ 96 to >99 per cent — always high, but not a calibrated absolute probability.

These limitations define the gap between “candidate” and “confirmed detection.” Closing them requires the follow-up programme described in Section 6.4.

7 CONCLUSIONS

We have presented an archival multi-wavelength analysis of HD 338492, a Gaia DR3 single-lined spectroscopic binary with a high mass function ($f(M) = 2.39 M_\odot$). Our principal findings are:

- (i) The mass function exceeds the Chandrasekhar limit from dynamics alone, independent of any stellar model.
- (ii) Adopting the SIMBAD B9 classification, the extinction-corrected primary mass is $M_1 = 4.40 \pm 1.32 M_\odot$, giving a minimum companion mass $M_{2,\text{min}} = 6.62 M_\odot$ — well above the neutron-star ceiling.
- (iii) No luminous companion is detected: a main-sequence star of the required mass would exceed the photometric detection threshold by a factor of 943. The companion is dark.
- (iv) Under fiducial prior assumptions, the Monte Carlo mass posterior gives $P(\text{BH}) = 99.7$ per cent. Sensitivity analysis (Table 3) shows this remains above ~ 96 per cent under B9-anchored configurations, dropping to ~ 89 per cent only if the B9 spectral classification is abandoned entirely. The value is model-dependent and should not be interpreted as a calibrated absolute probability.
- (v) Four of six alternative non-BH scenarios are excluded. The hierarchical triple is strongly constrained but not fully excluded; the stripped helium star is disfavoured but requires UV spectroscopy to rule out definitively.
- (vi) ROSAT and XMM non-detections are consistent with a dormant (non-accreting) compact object.
- (vii) **No independent radial-velocity confirmation exists.** The entire case rests on archival Gaia, photometric, and X-ray data, and the stellar characterisation depends on the SIMBAD B9 classification.

HD 338492 is a strong dormant black hole *candidate* that warrants urgent follow-up. We emphasise that the gap between candidate and confirmed detection is significant: independent RV verification of the orbit, a modern spectroscopic classification to replace the SIMBAD anchor, and UV observations to close the stripped-star scenario are all needed before the case can be considered settled. Ground-based radial-velocity monitoring (~ 8 – 10 epochs over ~ 90 d on a 2–4 m class telescope) is the single most impactful next step.

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DATA AVAILABILITY

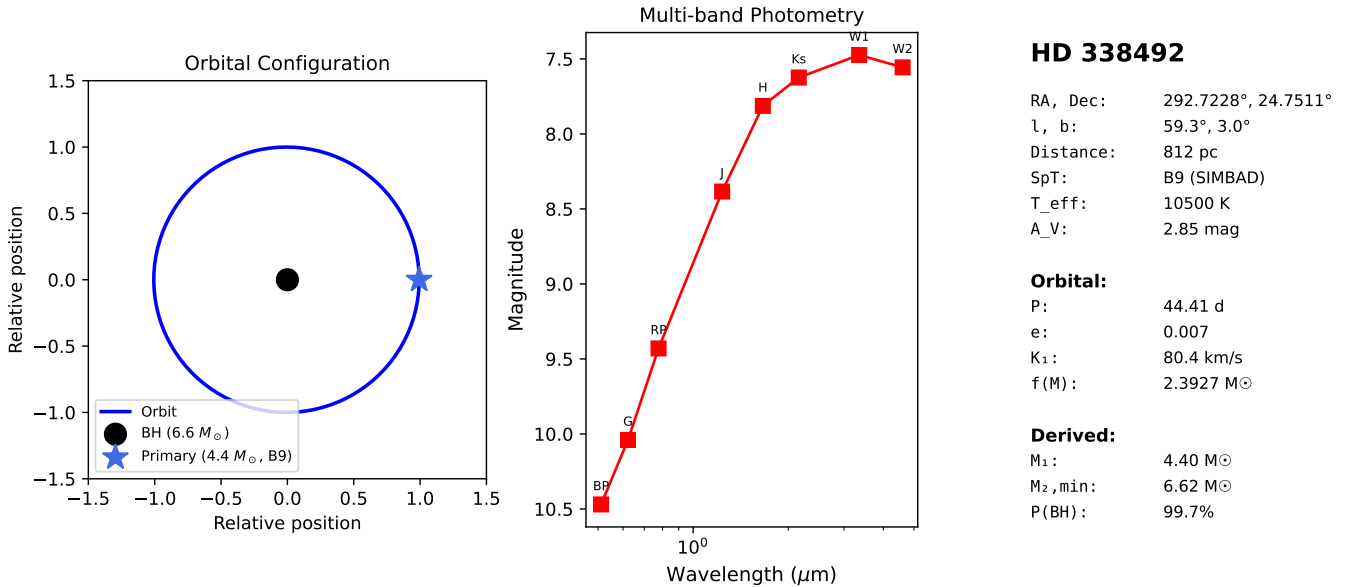
All observational data used in this paper are publicly available from the Gaia Archive (<https://gea.esac.esa.int/archive/>), VizieR, and SIMBAD. The analysis scripts, compiled input data, and machine-readable results are available at <https://github.com/jayalabaez/hd338492-black-hole-candidate>. Specifically: `02_fit_sed_extinction.py` produces Fig. 3; `03_compute_mass_posterior.py` produces Fig. 2; `04_companion_exclusion.py` produces Fig. 4; and `06_make_figures.py` produces Figs 1, 5, and 6. All numerical results reported in the text are reproduced in JSON files under `results/`. A frozen Zenodo release (DOI to be inserted at proof stage) will accompany the final version; in the interim, the full reproducibility package is available at the GitHub repository above.

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Table 1. Observed and derived properties of HD 338492.

Property	Value	Uncertainty	Source
<i>Astrometry (Gaia DR3)</i>			
RA (J2016.0)	292.72284°	—	Gaia DR3
Dec (J2016.0)	24.75113°	—	Gaia DR3
Parallax ϖ	1.232 mas	0.025	Gaia DR3
Distance	811.6 pc	—	$1/\varpi$
G	10.04 mag	0.003	Gaia DR3
G_{BP}	10.47 mag	0.003	Gaia DR3
G_{RP}	9.43 mag	0.004	Gaia DR3
RUWE	2.05	—	Gaia DR3
<i>NSS orbital solution (SB1)</i>			
Period P	44.41 d	0.01	Gaia NSS
Eccentricity e	0.007	0.003	Gaia NSS
Semi-amplitude K_1	80.355 km s ⁻¹	0.4	Gaia NSS
Significance σ	249.1	—	Gaia NSS
<i>Multi-wavelength photometry</i>			
J	8.384 mag	0.02	2MASS
H	7.813 mag	0.05	2MASS
K_s	7.624 mag	0.02	2MASS
$W1$	7.474 mag	0.03	AllWISE
$W2$	7.556 mag	0.02	AllWISE
$W3$	7.487 mag	0.02	AllWISE
$W4$	7.269 mag	0.07	AllWISE
<i>Derived quantities</i>			
Spectral type	B9	—	SIMBAD
T_{eff}	10 500 K	500	SpT $\rightarrow T_{\text{eff}}$
A_V	2.85 mag	0.3	Colour excess
$E(G_{BP} - G_{RP})$	1.189 mag	0.1	SpT anchor
M_G (corrected)	-1.71 mag	0.3	Derived
M_1 (primary mass)	4.40 M _⊙	1.32	HRD placement
$f(M)$ (mass function)	2.3927 M _⊙	0.04	Eq. (1)
$M_{2,\text{min}}$	6.62 M _⊙	—	Eq. (2)
$P(\text{BH})$	99.7%	—	MC posterior

HD 338492 — High-Mass Dark Companion Candidate**Figure 1.** Overview of the HD 338492 system. *Left:* Orbital configuration showing the nearly circular ($e = 0.007$) orbit. *Centre:* Multi-band observed photometry from Gaia, 2MASS, and WISE. *Right:* Summary of key system properties.

HD 338492 — Mass Constraints

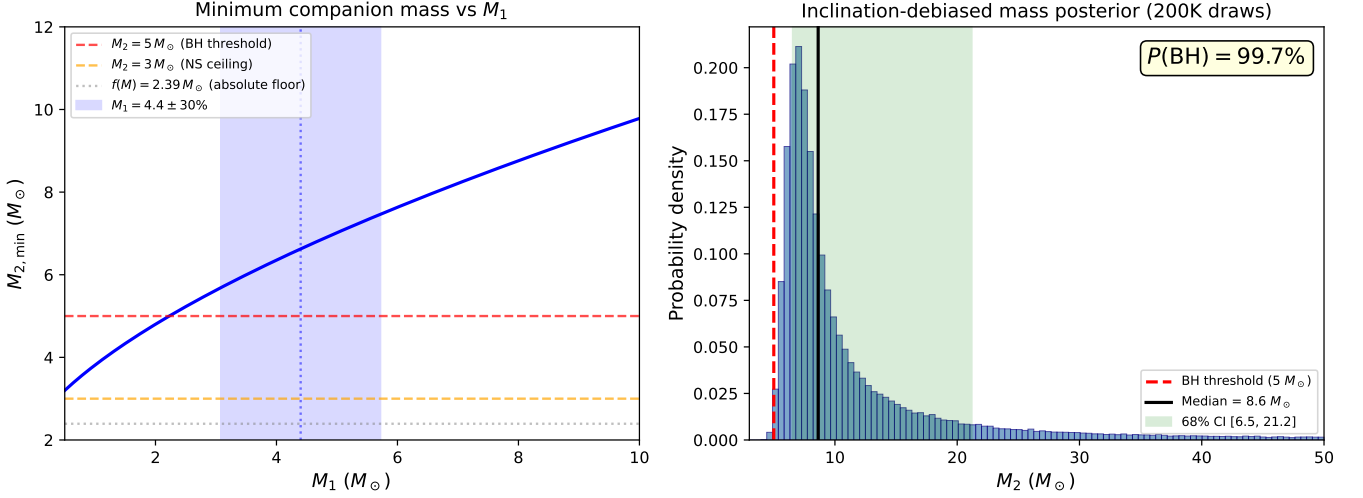


Figure 2. *Left:* Minimum companion mass $M_{2,\min}$ as a function of assumed primary mass M_1 . The dashed lines mark the adopted $M_1 = 4.40 M_\odot$ and the resulting $M_{2,\min} = 6.62 M_\odot$. *Bottom:* Monte Carlo posterior distribution of M_2 from 200 000 draws with isotropic inclination and log-normal M_1 priors. The shaded blue region marks $M_2 > 5 M_\odot$ (BH regime); the integrated probability is $P(\text{BH}) = 99.7$ per cent.

HD 338492 — SED Analysis

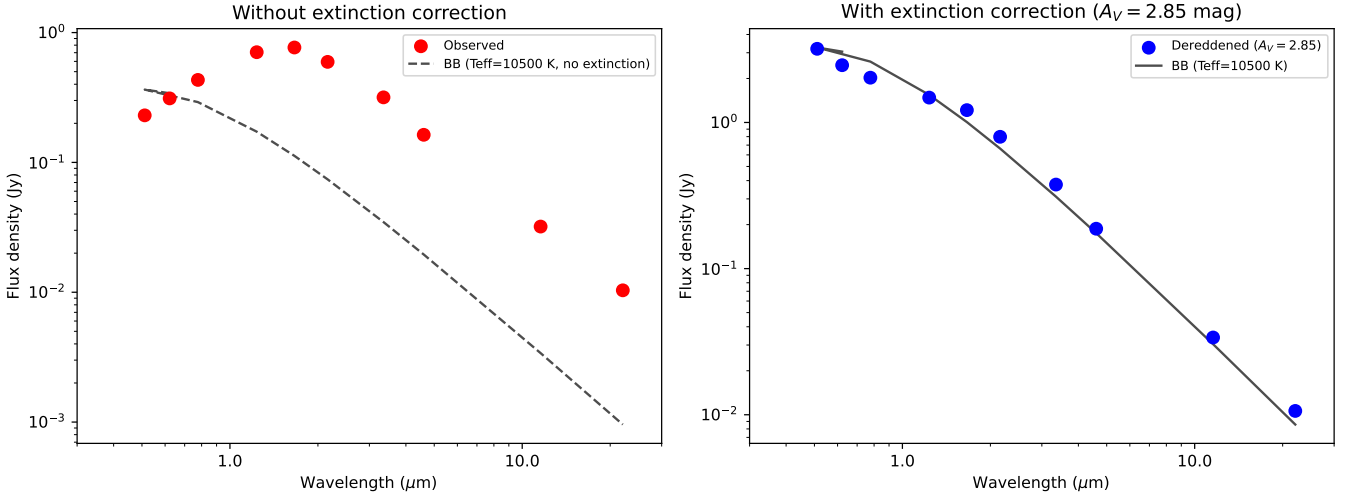


Figure 3. Spectral energy distribution of HD 338492. *Top:* Observed photometry compared to a 10 500 K blackbody — the SED is strongly reddened ($\chi^2 \approx 285$). *Bottom:* Dereddened photometry ($A_V = 2.85$) compared to the same model — $\chi^2 \approx 14$, a 20-fold improvement. No secondary SED component is required.

Table 2. Alternative scenario assessment.

Scenario	Verdict	Key constraint
MS companion	Excluded	943 $\times$ flux excess
White dwarf	Excluded	$f(M) > M_{\text{Ch}}$
Neutron star	Excluded	$M_{2,\min} = 2.9 M_{\text{TOV}}$
Triple system	Unlikely	Stability + photometry
Stripped He star	Disfavoured	No UV excess
Artefact	Excluded	$\sigma = 249.1$

Table 3. Sensitivity of $P(\text{BH})$ to prior assumptions. The fiducial configuration (row 3) adopts the SIMBAD B9 spectral anchor; rows below it explore wider M_1 priors, alternative extinction values, and the case where the B9 classification is abandoned entirely. All runs use 2×10^5 Monte Carlo draws with isotropic inclination priors.

Configuration	BH threshold ($M_\odot$)	M_1 ($M_\odot$)	σ_{M_1} ($M_\odot$)	A_V (mag)	$P(\text{BH})$ (%)	$\tilde{M}_2$ [68% CI] ($M_\odot$)
<i>Varying BH mass threshold</i>						
$M_2 > 3 M_\odot$ (conservative)	3	4.40	1.32	2.85	100.0	8.6 [6.5, 21.2]
$M_2 > 4 M_\odot$ (mass-gap edge)	4	4.40	1.32	2.85	100.0	8.6 [6.5, 21.2]
$M_2 > 5 M_\odot$ (fiducial)	5	4.40	1.32	2.85	99.7	8.6 [6.5, 21.2]
<i>Wider M_1 prior (50% fractional uncertainty)</i>						
Wide prior, $M_2 > 3$	3	4.40	2.20	2.85	100.0	8.7 [6.1, 21.3]
Wide prior, $M_2 > 5$	5	4.40	2.20	2.85	96.4	8.7 [6.1, 21.3]
<i>Varying extinction</i>						
Low A_V	5	3.96	1.19	2.3	99.2	8.3 [6.3, 20.6]
High A_V	5	4.84	1.45	3.4	99.9	9.0 [6.8, 21.7]
<i>Without B9 spectral anchor</i>						
Generic A/B star	5	3.00	1.50	2.85	88.7	7.5 [5.3, 19.4]

Table 4. Comparison of HD 338492 with the three confirmed Gaia dormant black hole systems. Gaia BH1, BH2 and BH3 values are from El-Badry et al. (2023a), El-Badry et al. (2023b) and Gaia Collaboration et al. (2024), respectively. HD 338492 remains at the candidate stage pending independent RV confirmation.

Property	Gaia BH1	Gaia BH2	Gaia BH3	HD 338492
P (d)	185.6	1277	11.6 yr	44.41
e	0.45	0.52	0.73	0.007
K_1 (km s^{-1})	65.9	4.6	4.5	80.4
$f(M)$ ($M_\odot$)	3.96	0.37	3.57	2.39
M_1 ($M_\odot$)	0.93	1.07	0.76	4.40
M_2 ($M_\odot$)	9.62	8.94	32.70	>6.62
$P(\text{BH})$	—	—	—	99.7%
Status	Confirmed	Confirmed	Confirmed	Candidate

HD 338492 — Luminous Companion Exclusion

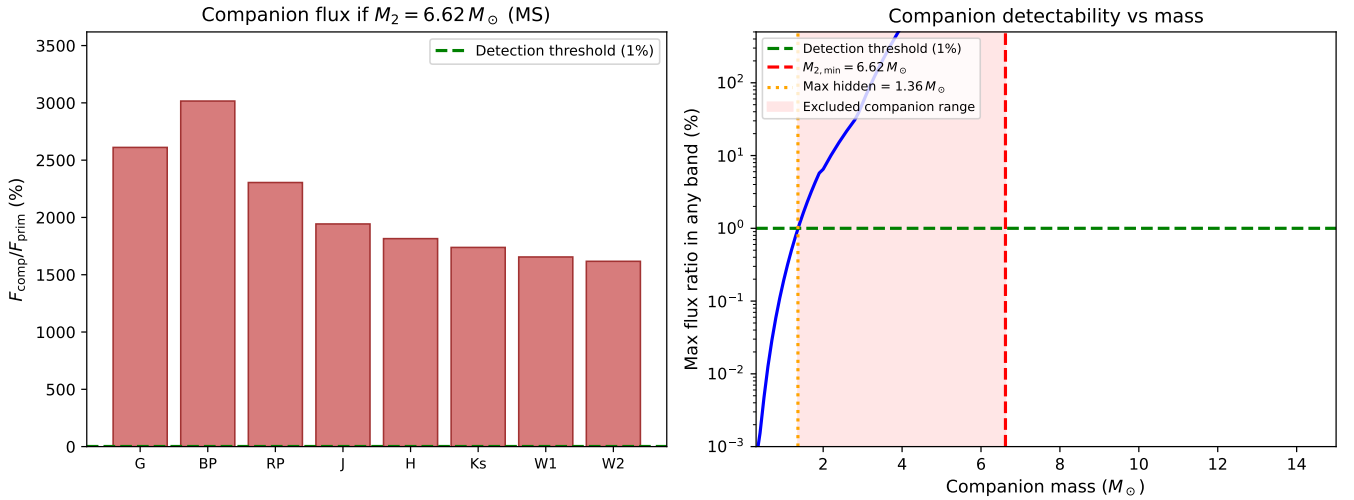


Figure 4. *Left:* Expected companion-to-primary flux ratio in each photometric band for a hypothetical $6.62 M_\odot$ MS companion. All bands exceed the 1 per cent detection threshold. *Right:* Maximum flux ratio as a function of companion mass. The detectable mass threshold is $\sim 1.4 M_\odot$, far below $M_{2,\text{min}} = 6.62 M_\odot$.

HD 338492 — Stellar classification

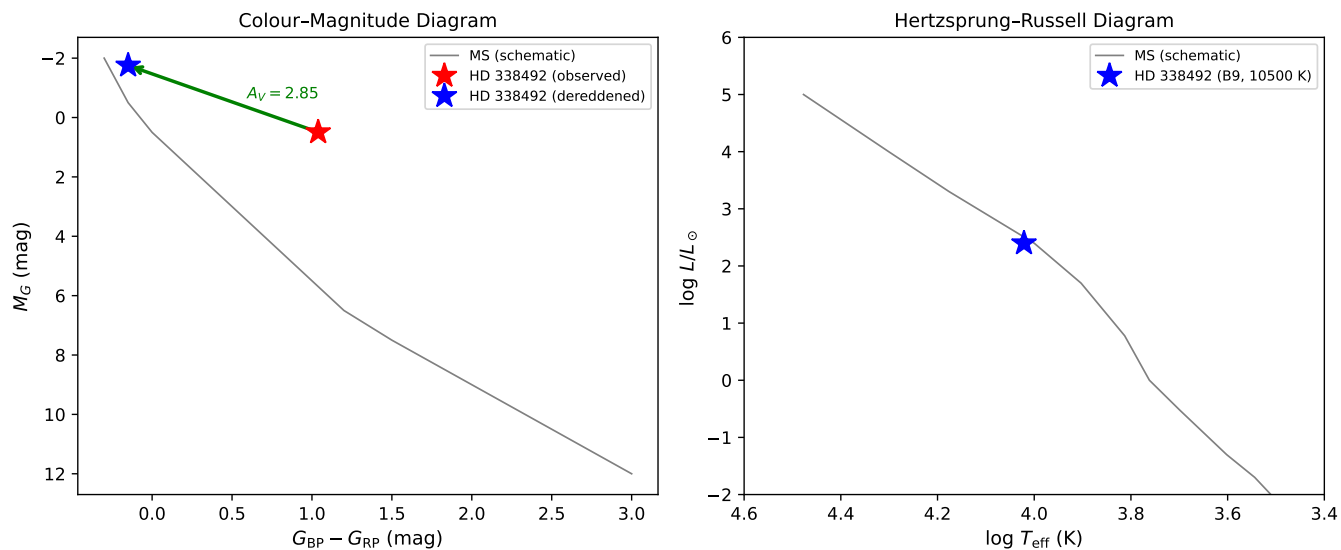


Figure 5. *Left:* Colour-magnitude diagram showing HD 338492 before (red) and after (blue) extinction correction. The green arrow indicates the reddening vector ($A_V = 2.85$). *Right:* Hertzsprung-Russell diagram placement. The dereddened position is consistent with a B9 main-sequence star of $\sim 4.4 M_\odot$.

HD 338492 — Confirmation Checklist

Check	Result	Detail
C1: Gaia quality		RUWE=2.05, σ =249.1
C2: Orbit solution		P=44.41d, e=0.007, K ₁ =80.4
C3: Mass function		f(M)=2.39 > Chandrasekhar
C4: Spectral type		B9 → Teff=10500 K
C5: HR diagram		M ₁ =4.40 on MS
C6: Extinction		A _V =2.85 (SpT-dependent)
C7: SED fit		χ^2 =14.1 (BB approx)
C8: Companion light		943× exclusion
C9: X-ray		Non-detection (dormant)
C10: UV excess		No GALEX coverage
C11: Mass posterior		P(BH)=99.7%
C12: Alt. scenarios		2/6 not fully excluded
C13: Independent RV		Not yet obtained

Score: 7P / 4O / 2F — Verdict: STRONG BH CANDIDATE (requires RV confirmation)

Figure 6. Confirmation checklist for HD 338492, showing the outcome of each diagnostic test. Green circles indicate PASS, orange indicates OPEN (inconclusive), red indicates FAIL (not yet satisfied). The system scores 7P/4O/2F — a “compelling” BH candidate requiring follow-up to close the two remaining failures (independent RV and full scenario exclusion).